COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 4 : SLO 1 — Shift-and-Add Unsigned Multiplication
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1. Vocabulary Match-Up (Remembering)

• 1. Multiplicand → C. The number that is being multiplied
• 2. Multiplier → D. The number by which you multiply
• 3. Partial Product → E. Result of multiplying the multiplicand by a single multiplier bit
• 4. Accumulator → B. Register that holds the intermediate product
• 5. Shift-Left → F. Moves all bits one position to the left
• 6. Product → A. The result of the multiplication

2. Step-by-Step Description (Understanding) — correct order

• 1 → Initialize the Accumulator to all zeros.
• 2 → If the LSB of the Multiplier is 1, add the Multiplicand to the Accumulator.
• 3 → Shift the Multiplicand one position to the left.
• 4 → Shift the Multiplier one position to the right.
• (In hardware, steps 2–4 form one cycle repeated for each multiplier bit.)

3. Applying the Concept (Applying): 1101 (13) × 1011 (11) by shift-and-add

• Initial: Multiplicand = 0000 1101, Multiplier = 1011, Accumulator = 0000 0000
• Iteration 1: LSB = 1 → Add: Accumulator = 0000 1101; then shift → Mcand = 0001 1010, Mplier = 0101
• Iteration 2: LSB = 1 → Add shifted multiplicand: 0000 1101 + 0001 1010 = 0010 0111 (Accumulator = 0010 0111); shift → Mcand = 0011 0100, Mplier = 0010
• Iteration 3: LSB = 0 → No add; shift → Mcand = 0110 1000, Mplier = 0001
• Iteration 4: LSB = 1 → Add shifted multiplicand: 0010 0111 + 0110 1000 = 1000 1111
• Final Binary Product: 10001111 Final Decimal Product: 143 (13 × 11 = 143)

4. Error Analysis (Analyzing): 101 × 110

• Incorrect step number: Step 3.
• Error: in Step 3 the accumulator holds 001010 and the shifted multiplicand is 010100; since the multiplier LSB is 1 the sum must be 001010 + 010100 = 011110. The student wrote 011100 — a binary addition error.

5. Design a Problem (Creating)

• Chosen 3-bit multiplicand: 111 (7) Chosen 3-bit multiplier: 101 (5)
• Step 1: LSB = 1 → add → Accumulator = 000111; shift → Mcand = 001110, Mplier = 010
• Step 2: LSB = 0 → no add; shift → Mcand = 011100, Mplier = 001
• Step 3: LSB = 1 → add → 000111 + 011100 = 100011; shift → done
• Final Answer: 100011 (binary) = 35 (decimal)